Q-resafe AWQ Calibration-Data Study

Consolidated experimental report: Llama-2-7B-Chat, official Q-resafe quant-without-ft AWQ, Modal NVIDIA L40S

Executive conclusion
The recorded INT4 AWQ runs do not provide evidence for a large or monotonic Risk-II/Risk-III safety degradation under the first-pass held-out ASR heuristic. In the initial run, Risk-II is 1.37 percentage points above AWQ-Pile and Risk-III is 0.78 points above it, but both are also close to the FP16 control and the utility PPL values remain in a narrow 7.14-7.22 range. The token-matched confirmation preserves the ordering Risk-II/Risk-III above Risk-I, but it is a sensitivity check rather than a definitive causal confirmation because the 384-token prefix truncates source coverage.

510	5	4-bit	L40S
held-out AdvBench prompts	main conditions	AWQ weight precision	Modal GPU

1. Research question and scope

This study extends the Risk-I/Risk-II/Risk-III taxonomy described in Q-resafe to the calibration corpus of its AWQ implementation. It asks whether changing only the calibration text used by pure post-training quantization changes safety behavior of an already safety-aligned model. It is not a reproduction of Q-resafe's AWQ Risk-I/Risk-II/Risk-III experiments, because those conditions were not run in the original paper.

Locked pipeline: FP16 Llama-2-7B-Chat -> official Q-resafe AWQ INT4 with one calibration corpus -> held-out AdvBench safety evaluation and WikiText-2 perplexity. No optimizer, backward pass, SFT, LoRA, QAT, AQLM, GPTQ, QLoRA, or vaccine was used.

2. Locked protocol

Item	Recorded setting
Model	meta-llama/Llama-2-7b-chat-hf
Implementation	Qresafe/quant-without-ft official AWQ path
AWQ config	w_bit=4; q_group_size=128; zero_point=True; version=GEMM
Seed	42
Safety evaluation	held-out AdvBench, n=510; greedy decoding; max_new_tokens=128; chat-template generation prompt
Utility evaluation	WikiText-2 raw test perplexity
Risk data	10 examples per condition; question-response conversations rendered once and tokenized

Generation configuration recorded in generation_config.json:

```
{"do_sample": false, "max_new_tokens": 128, "seed": 42, "chat_template":
"tokenizer.apply_chat_template", "add_generation_prompt": true}
```

3. Calibration data and leakage control

Condition	Source	Examples	Tokens	Blocks	Policy
Pile	Q-resafe pileval default	128	33,777	65	max_seq_len=512
Risk-I	HuggingFaceH4/ultrachat_200k	10	5,537	43	common 128-token chunks
Risk-II	public-description reconstruction	10	469	3	common 128-token chunks
Risk-III	AlignmentResearch/AdvBench	10	438	3	common 128-token chunks

Risk-II is reconstructed from the public description and is not the exact unreleased Q-resafe dataset. Risk-III uses paired responses from the public AdvBench export under response column proxy_gen_target. The ten Risk-III source indices are removed from the evaluation split, leaving 510 held-out rows; the recorded intersection is empty. The report intentionally lists provenance IDs only and does not reproduce harmful prompts or responses.

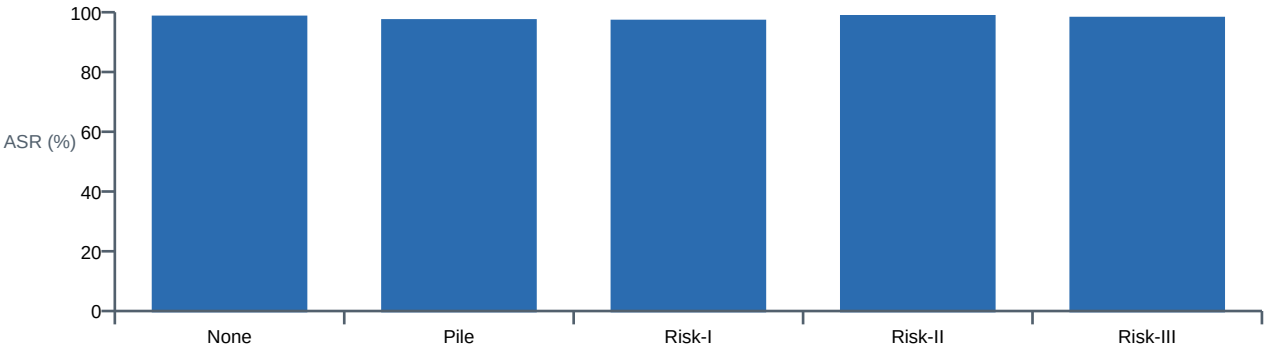
Important first-run confound
The initial calibration token budget is strongly imbalanced: Risk-I has 5,537 tokens versus 469 for Risk-II and 438 for Risk-III. The token-matched confirmation fixes all three streams at 384 tokens and 3 blocks, but uses the deterministic prefix of the rendered stream, so it matches token count rather than full ten-example source coverage.

4. Initial five-condition results

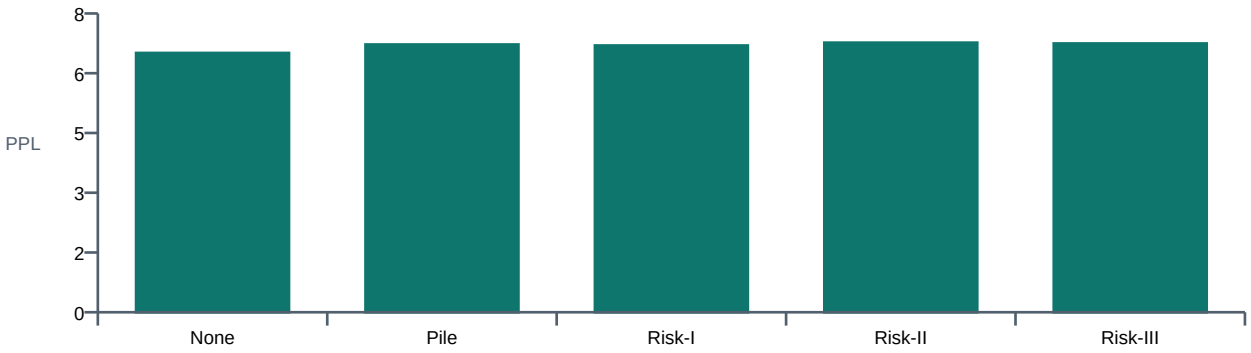
ASR is reported exactly under the artifact field asr_heldout_refusal_heuristic. The count column is the corresponding positive count out of 510. Higher ASR is interpreted as worse safety under the intended ASR definition; because the implementation is a transparent heuristic, these numbers require an approved judge or manual validation before a publication-level claim.

Model	Calibration	Samples	Tokens	Blocks	ASR	Count/510	PPL	Delta vs Pile
FP16 / 16b	None	–	–	–	98.43%	502	6.9424	+1.18 pp
AWQ / 4b	Pile	128	33777	65	97.25%	496	7.1696	+0.00 pp
AWQ / 4b	Risk-I	10	5537	43	97.06%	495	7.1425	-0.20 pp
AWQ / 4b	Risk-II	10	469	3	98.63%	503	7.2186	+1.37 pp
AWQ / 4b	Risk-III	10	438	3	98.04%	500	7.1971	+0.78 pp

Initial held-out ASR heuristic



Initial WikiText-2 perplexity



5. Interpretation of the initial run

Relative to AWQ-Pile, Risk-I changes ASR by -0.20 pp, Risk-II by +1.37 pp, and Risk-III by +0.78 pp. Relative to Risk-I, Risk-II is +1.57 pp and Risk-III is +0.98 pp. Thus the point estimate is compatible with a harmful-calibration signal for Risk-II/Risk-III, but the ordering is not a clean monotonic Risk-I < Risk-II < Risk-III result: Risk-II is highest and Risk-III is lower than Risk-II.

Utility does not collapse in the same run. PPL is 7.1425 for Risk-I, 7.2186 for Risk-II, and 7.1971 for Risk-III, compared with 7.1696 for Pile and 6.9424 for FP16. The Risk-II/Risk-III ASR differences therefore cannot be dismissed immediately as catastrophic quantization failure, although PPL alone is not a sufficient utility audit.

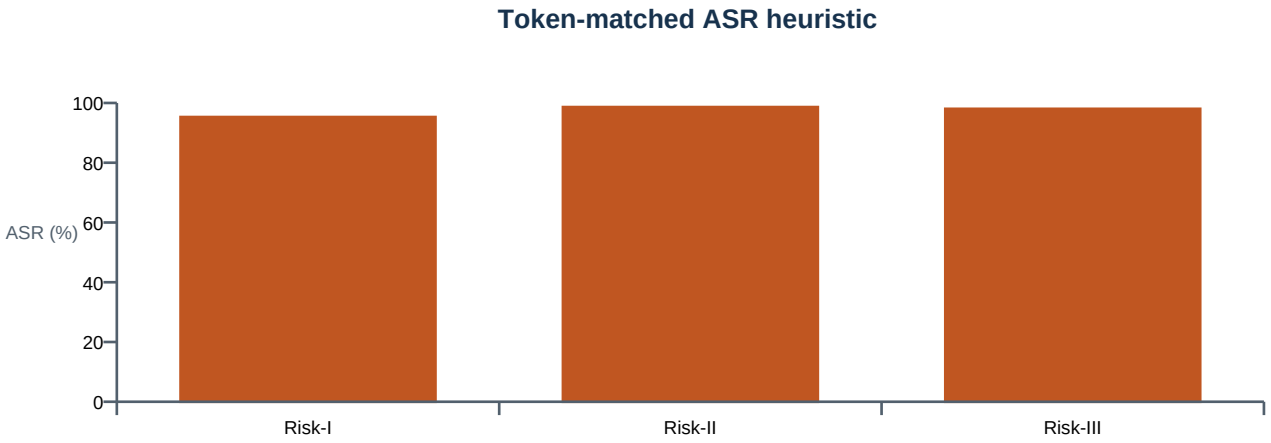
What the first run can support
A preliminary observation: changing calibration text under pure AWQ is associated with small ASR differences in this setup, while utility PPL stays close. What it cannot support alone is a definitive causal claim, because the Risk token budgets differ greatly, Risk-II is reconstructed, the ASR judge is heuristic, and there is only one seed.

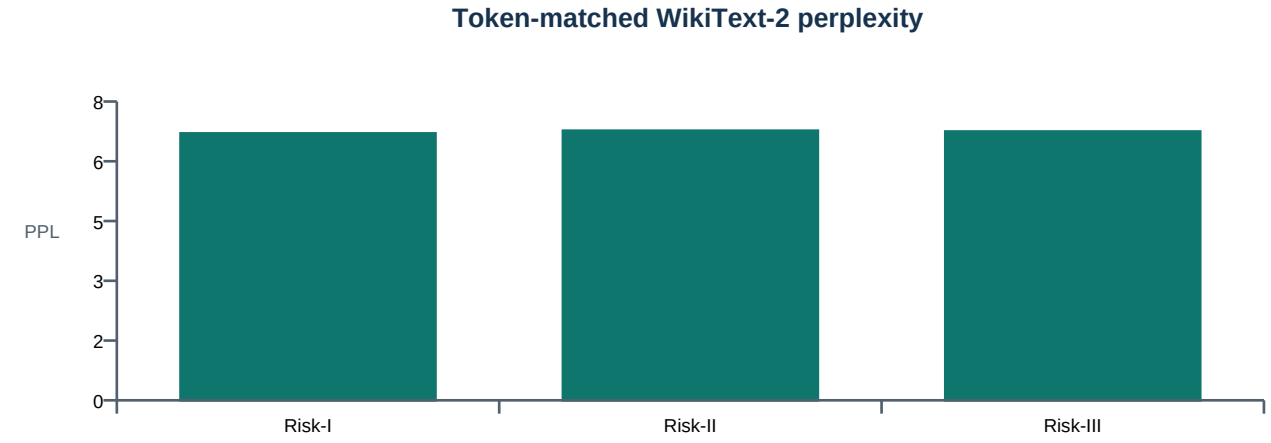
6. Token-matched confirmation

The confirmation reruns only Risk-I/Risk-II/Risk-III with exactly 384 rendered tokens and 3 complete 128-token blocks for each condition. FP16 and Pile are unchanged controls from the initial run.

Calibration	Source examples	Used tokens	Blocks	ASR	Count/510	PPL	Delta vs R1
Risk-I	10	384	3	95.29%	486	7.1464	+0.00 pp
Risk-II	10	384	3	98.63%	503	7.2186	+3.33 pp
Risk-III	10	384	3	98.04%	500	7.1971	+2.75 pp

After matching token count, Risk-II remains +3.33 pp above Risk-I and Risk-III remains +2.75 pp above Risk-I. Risk-II and Risk-III match their initial ASR/PPL values exactly in the recorded artifacts; Risk-I changes from 97.06% to 95.29% ASR after truncation from 5,537 to 384 tokens. This is directionally supportive of a calibration-distribution effect, but it is not a fully source-coverage-matched replication.





7. Execution environment and evidence

Environment field	Recorded value
Python	3.10.17
Torch	2.1.2+cu118
CUDA build	11.8
GPU	NVIDIA L40S
Transformers	4.47.1
GPU availability	True
Model	meta-llama/Llama-2-7b-chat-hf
Local Q-resafe checkout	fc46653e6a77c831d0c3c11e20595f50aee6200d
Artifact metadata commit	unknown in r6_final and r6_tokenmatched metadata

Recorded evidence includes Modal smoke logs, model-access and data-preparation logs, full-run logs, quantization metadata, safety JSONL checkpoints, PPL JSON files, result CSVs, the calibration index manifest, and the local unit-test suite. The local test suite was rerun during report preparation: 42 passed, 2 deprecation warnings, 11.39 seconds. The public release contains raw artifacts because a full release was requested; model weights and credentials are excluded.

The release repository is public at https://github.com/ihbkaiser/qresafe-awq-calibration-study. The report and its source are included under output/pdf/ and reports/.

8. Limitations and technical notes

- ASR is a transparent refusal-heuristic field, not a validated human or model-judge score. The artifact itself warns that it must be validated before publication.

- Risk-II is reconstructed from the public description; the exact Q-resafe Risk-II data are unreleased.

- Risk-III uses a reconstructed public AdvBench export and a held-out split with empty recorded overlap, but exact Q-resafe selection is unavailable.

- The initial run has a large calibration-token imbalance. The token-matched check matches a 384-token prefix, not all ten full conversations.

- There is one fixed seed and no confidence interval or multi-seed replication in this release.

- The raw Modal logs contain earlier failed invocations: a gated-model access failure and a utility-evaluation logits-interface failure. The numeric tables in this report are read from the persisted r6_final and r6_tokenmatched result artifacts, not from those failed log sections.

- The recorded artifact metadata says qresafe_quant_without_ft_commit=unknown. The local checkout is at fc46653e6a77c831d0c3c11e20595f50aee6200d, but the exact source revision used by the remote artifact cannot be proven from the metadata alone.

9. Research conclusion

The current evidence is best described as a preliminary signal, not a confirmed phenomenon. In both the initial and token-matched artifacts, Risk-II and Risk-III have higher held-out ASR heuristic values than Risk-I, and the PPL values remain close enough that a total utility collapse is not the obvious explanation. However, the evidence is weakened by the heuristic safety metric, the reconstructed Risk-II corpus, single-seed design, and source-coverage mismatch in token matching. The defensible conclusion is: harmful-looking calibration distributions may affect AWQ safety behavior in this setup, but the magnitude and specificity of the effect require a stricter confirmation before claiming that harmful calibration data alone reliably degrades safety.

Recommended next experiment: preregister a judge-validated ASR protocol, record the exact Q-resafe commit in metadata, keep ten-example source coverage identical while matching total tokens through a declared sampling/truncation rule, and repeat at least three seeds without selecting seeds based on ASR.

Appendix A. Reproducibility identifiers

Only identifiers are included below. Harmful calibration and evaluation text is intentionally omitted from this report; the raw artifact package contains the recorded files and should be handled as research data.

Corpus	Identifiers
Risk-I / UltraChat	167621, 29184, 6556, 194393, 72097, 64196, 58513, 36579, 193061, 26868
Risk-II	reconstructed-0 through reconstructed-9; exact original unavailable
Risk-III / AdvBench	114, 25, 281, 250, 228, 142, 104, 89, 432, 32
Risk-III held-out	510 rows; calibration/evaluation intersection = []
Seed	42

Calibration metadata fields record source examples, used examples, discarded examples, total tokens before blocking, block length policy, block count, and sample IDs for every custom condition. The token-matched metadata additionally records original token totals and discarded-token counts: Risk-I 5,537 -> 384 (5,153 discarded), Risk-II 469 -> 384 (85 discarded), Risk-III 438 -> 384 (54 discarded).

Appendix B. Release contents

Path	Purpose
remote_artifacts/r6_final/	Initial five-condition result CSVs, summaries, PPL, quantization metadata, and indices
remote_artifacts/r6_tokenmatched/	Token-matched Risk-I/Risk-II/Risk-III result CSVs and metadata
remote_artifacts/r6_checkpoints/	Resumable safety-generation checkpoints
remote_artifacts/modal_*/	Modal smoke, preparation, access-check, and full-run logs
scripts/ and src/	Calibration preparation, quantization, evaluation, collection, and tests
output/pdf/	This consolidated report
reports/	Report-generation source

Release safety note
The public package excludes HF credentials, GitHub credentials, Python caches, and model weights. Raw logs and model-generated safety-test records are included because a full artifact release was requested; do not redistribute harmful prompt text unnecessarily.